Lost in Translation: Gender Erasure in English-to-Hindi Machine Translation

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37,345 Test instances

5 Translation models

63.2-76.3% Ergative usage

83.8% Human-classifier agreement

Problem Statement

English-to-Hindi MT must realize grammatical gender in Hindi even when English provides weak cues. Current systems often remain fluent but erase gender through ergative constructions or inject masculine defaults into neutral contexts.

Research Gaps

- Prior bias audits focus mainly on European MT or broad multilingual benchmarks.
- Hindi work does not isolate English-to-Hindi ergative neutralization as the core failure mechanism in a large controlled benchmark.
- Prompting-based mitigation for Hindi remains underexplored, especially under structurally gender-erasing outputs.

Experimental Setup

- Dataset: 12,677 male · 12,963 female · 11,705 neutral
- Cleaned setup: 0 unresolved predictions across all 5 models
- Prompting: Sarvam baseline + 4 instruction variants (L1–L4)
- Human eval: 100 doubly annotated items · κ = 0.776

Methodology

- Build a 37,345-instance benchmark across 12 linguistically motivated categories.
- Translate each instance with 5 systems: Sarvam, Helsinki, NLLB, IndicTrans2, GPT-4o-mini.
- Classify outputs with a grammar-aware pipeline: lexical rules, morphology, ergative-aware rules plus unified GPT-4o-mini fallback.
- Measure accuracy, masculine default rate, ergative rate, per-category performance, and human agreement.

Evaluation Pipeline

STAGE 01
Benchmark
Controlled English
1. 37,345 sentences
2. 12 categories
3. 45 professions
4. 30 names
5. 50 profession pairs

STAGE 02
Translation
EN → HI
1. Sarvam
2. Helsinki Opus-MT
3. NLLB-200
4. IndicTrans2
5. GPT-4o-mini
6. Sarvam prompting variants

STAGE 03
Classifier
Grammar-aware
1. Explicit gender words
2. Name detection
3. Verb agreement rules
4. Ergative handling
5. GPT-4o-mini fallback

STAGE 04
Evaluation
Metrics & tests
1. Accuracy
2. Masculine default rate
3. Per-category breakdown
4. Bootstrap CIs
5. Permutation tests

63.2 to 76.3% of outputs use ergative ने constructions that can hide subject gender.

ENGLISH SOURCE — TRANSLATED HINDI OUTPUT

He completed the report. → उसने रिपोर्ट पूरी की।

In many Hindi perfective sentences the subject takes ने. The verb then agrees with the object, not the subject. उसने can mean *he* or *she*, and की agrees with रिपोर्ट rather than the person. *The gender signal is **neutralized** by the target grammar. The classifier reports neutral.*

MALE → NEUTRAL

Failure modes

These translations stay fluent, but the gender cue disappears. The recurring shift is not female → male, but female → neutral

FEMALE → NEUTRAL

MODE 01 · EXPLICIT GENDER
Copula "है" carries no gender
"She is a skilled engineer."
वह एक कुशल इंजीनियर है।
Hindi copula "है" does not mark gender. The female pronoun is lost.

MODE 02 · LATE BINDING
Referent resolves after ने
"The engineer reviewed all documents. Afterward, she signed them."
इंजीनियर ने सारे कागज़ों की समीक्षा की। उसके बाद उसने उन पर हस्ताक्षर कर दिए।
"उसने" in the second sentence does not reveal subject gender.

MODE 03 · WINOGRAD COREFERENCE
Pronoun-antecedent collapsed
"The engineer called the nurse since she required assistance."
इंजीनियर ने नर्स को फोन किया क्योंकि उसे मदद की आवश्यकता थी।
Pronoun cue maps to gender-neutral "उसे". The coreference is unresolved.

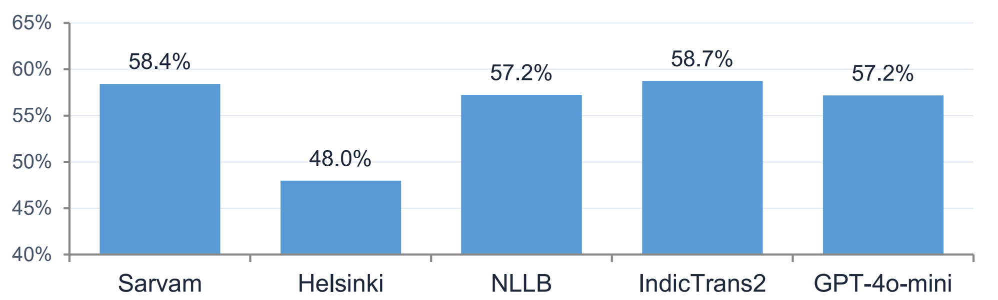
HUMAN VALIDATION

93.9% Agreement on neutral predictions

κ = 0.776 Cohen's κ on gender labels

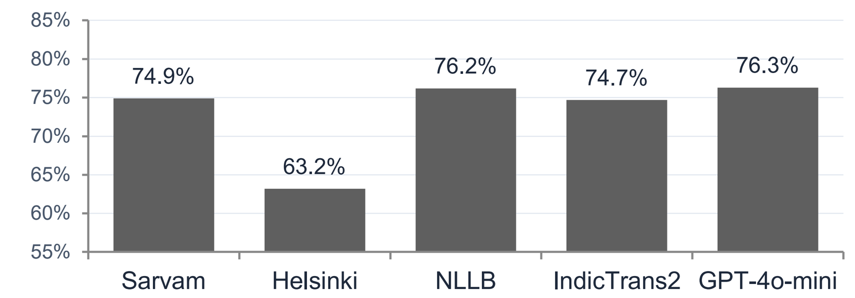
Main Results

Overall Accuracy Across Five Models



IndicTrans2 leads, but all 5 systems show substantial gender loss.

Gender Loss Tracks Ergative Usage

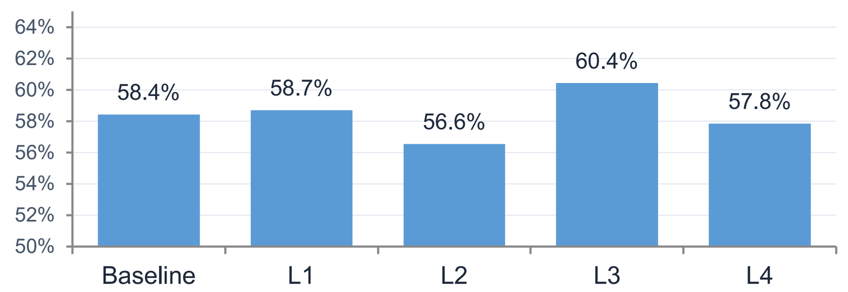


High fluency does not imply gender preservation.

Conclusions

- English-to-Hindi MT frequently erases gender through ergative constructions, not random failures.
- The pattern holds across 5 models - both specialized MT systems and a general-purpose LLM.
- Prompting improves overt defaults but does not solve the structural mechanism of gender loss.

Sarvam Prompting Variants



L3 is best, but prompting alone does not fix ergative neutralization.

Name Cues Survive; Pronoun/Coreference Cues Collapse

	Sarvam	Helsinki	NLLB	IndicTrans2
Name only	97.7	39.8	94.7	98.0
Name + profession	80.6	48.4	78.3	84.4
Neutral profession	94.4	80.0	99.2	96.7
Explicit gender	26.7	32.9	24.3	21.7
Late binding	7.8	5.6	0.1	7.2
Winograd coref	5.7	14.7	1.5	6.7

Name cues remain strong; pronoun and coreference cues collapse.

Limitations and Future Work

- Benchmark is controlled and diagnostic; validation on naturalistic, longer, real-world English-to-Hindi inputs is still needed.
- Human evaluation is promising but modest in scale and should be expanded.
- Future work should test constrained decoding, re-ranking, and gender-aware decoding to preserve gender without reducing fluency.

Central Thesis

The main failure is gender neutralization across explicit gender, late binding, and Winograd coreference. Ergative constructions using ने are the main mechanism behind this neutralization, not the bottleneck itself.

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